Shih-Kai Chou (S.-K. Chou)

Publications

S.-K. Chou, J.-D. Chai, and H.-S. Goan, “Efficient active space construction for quantum computing via Thermally-Assisted-Occupation-Density Functional Theory”, In preparation

- Implement an efficient quantum embedding approach

C.-T. Chu, S.-K. Chou, and H.-S. Goan, “Demonstration of quantum computation of molecular properties in agreement with experimental data on NISQ devices”, In preparation

- Use quantum machine learning to predict potential energy curves

S.-K. Chou, J.-P. Chou, A. Hu, Y.-C. Cheng, and H.-S. Goan, “Accurate harmonic vibrational frequencies for diatomic molecules via quantum computing”, Phys. Rev. Res. **5**, 043216 (2023).

Projects

Y.-H. Tung, S.-K. Chou, J.-Y. Huang, J.-P. Chou, Y.-C. Cheng, and H.-S. Goan, “A practical quantum computing approach to calculating protein–drug binding energies”, Poster at JSQC 2025

- Use quantum computation to calculate the protein–ligand binding energy

R.-Y. Chen, S.-K. Chou, J.-Y. Huang, Y.-C. Cheng, H.-S. Goan, and IBM Quantum, “Quantum-centric simulations using Daubechies wavelet basis set”, Poster at JSQC 2025

- The Road to Quantum Utility Program 2025 (collaborate with IBM Quantum team)

Conference Presentations

S.-K. Chou, *Utilizing TAO-DFT orbitals for active space construction in quantum computational chemistry*, Joint Symposium on Quantum Computing, Keio University, Tokyo, Japan, 2025.

S.-K. Chou, *Accurate and efficient quantum computation of molecular properties*, Joint Symposium on Quantum Computing, Yonsei University, Seoul, Korea, 2023.